Radiacode Track Analyzer: open-source analysis of GPS-tagged radiation tracks from consumer CsI(Tl) dosimeters

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Abstract

The RadiaCode 100-series is a palm-sized CsI(Tl) scintillation dosimeter whose companion application logs GPS-tagged tracks of count rate and dose rate. The application maps those tracks but does not analyse them, and the file format is undocumented. *Radiacode Track Analyzer* (`rcta`) is an MIT-licensed Python package — a headless command-line tool plus an optional Qt desktop application — that turns such a track into a quantitative result with its uncertainty attached. It reports a Currie minimum detectable dose rate whenever it finds no detection, corrects a multi-scale scan for its $\sim 2\times10^4$ implicit tests, de-weights correlations on a series whose lag-1 autocorrelation is 0.9985, separates GPS jitter from travelled distance, and validates imported aircraft altitude against the detector's own fixes before use. We also give a reverse-engineered specification of the format.

1 Introduction

Inexpensive scintillation detectors have put mobile radiation measurement within reach of undergraduate laboratories, environmental surveys and citizen-science groups. The RadiaCode 100-series is representative: a palm-sized CsI(Tl) instrument logging count rate and dose rate tagged with GPS position and horizontal accuracy, so a walk, a drive or a flight each produce a single file. What the vendor application does with that file is draw it. A heat-mapped track is persuasive, and a coloured line reads as signal whether or not one is present; the step to a defensible statement is left to the user, and the format is undocumented. `rcta` addresses both gaps, on the premise that the useful output of a mobile survey is usually a bound, not a detection.

2 Implementation

`rcta` is pure Python (≥ 3.10) requiring only NumPy and SciPy; PySide6 and pyqtgraph are an optional extra, so the analysis core runs headless or in a notebook. The package separates `core/` (pure NumPy/SciPy, never imports Qt, never touches the network) from `net/` and `gui/`, and those boundaries are enforced by a test module that inspects the import graph. The suite is 863 tests against ∼12,600 lines, and every quantitative claim below is pinned by a regression test.

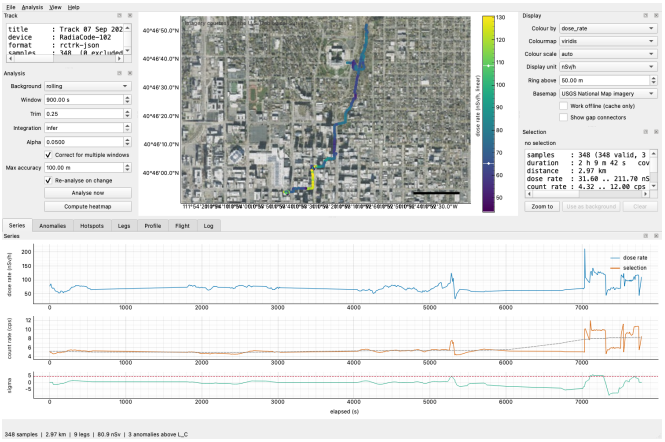


Figure 1: A 2 h walking survey: 348 samples, 2.97 km, background 5.31 cps (70.1 nSv/h). One feature reaches $7.4\,\sigma$ on two independent passes; a second, seen once, is only *possible* at $4.6\,\sigma$.

```
rcta-cli info    TRACK.rctrk
rcta-cli analyze TRACK.rctrk --json report.json
rcta-cli flight  TRACK.rctrk --flight DL2618 \
                 --date 2026-09-07
rcta                          # desktop application
```

Two real acquisitions ship with the package — a 2 h walking survey (Fig. 1) and a 2 h 42 m flight — so worked examples include the awkward parts: dropouts, out-of-order samples, 300 m fixes. The vendor publishes no specification: the application writes two dialects under one extension, randomises key order within a record, emits samples out of time order, and records dose rate in microroentgen per hour without recording the unit. `docs/FORMAT.md` is a specification measured against real files and enforced by tests; the unit was settled against a walking log and the application's own display, after an earlier parser read the values as nSv/h and was wrong tenfold.

3 Statistical treatment

Detection limits. When nothing exceeds the critical level L_C, `rcta` reports the minimum detectable dose rate against integration window in place of a null verdict — on the bundled flight, 1160 nSv/h in a single 2 s sample against 133 nSv/h over 128 s. This makes the Currie framework [1] concrete for a moving detector and exposes the $\sqrt{t}$ trade-off between sensitivity and spatial resolution. The same data show the energy-dependent dose response [2]: 13 nSv/h per cps on the ground against 51 at cruise, so the ratio is computed per track rather than assumed.

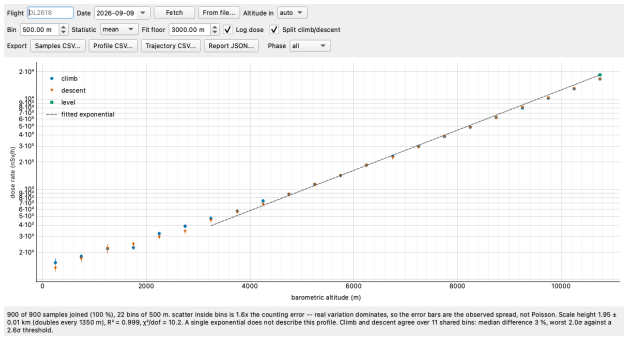


Figure 2: Cosmic dose rate against altitude, log ordinate, climb and descent separate. A constant scale height is a straight line; bins below 3 km bend away because terrestrial gamma is altitude-independent, hence the fit floor.

Multiplicity. Seven window widths across 2,964 samples are 20,748 implicit tests. At $\alpha = 0.05$ the uncorrected $1.64\,\sigma$ threshold returns of order a thousand "detections" from clean data; Bonferroni moves it to $4.57\,\sigma$ and the count to zero, verified over forty synthetic background tracks.

Serial dependence. The count rate has a lag-1 autocorrelation of 0.9985, so correlations carry a Bartlett effective sample size [3]. For speed against dose rate on the walk, $n = 347$ becomes $n_{\text{eff}} = 153$ and $p = 0.073$ becomes $p = 0.236$; on the flight the naive and corrected p-values differ by 260 orders of magnitude. The tool also states what no p-value can: on foot, slowing near something of interest couples speed to dwell time, so a surviving correlation is confounded.

Estimator and geometry. Background is a symmetrically trimmed mean of a rolling window, not a low quantile: the 0.25 quantile of an 18 cps Poisson background sits 2.0 cps below the true mean, which over a 64-sample window is $5.3\,\sigma$ of detection manufactured by the estimator. Geometry misleads similarly. At 1.4 m/s with 2 s logging each step is 2.8 m against an 8 m fix, so raw summed GPS distance overstates a walk by 5.2× against known truth; smoothing within an automatically selected window lands within 3 % across 3–20 m accuracy. Interpolated surfaces use Nadaraya–Watson [4] with two masks, since a one-dimensional path cannot support a filled map. Accumulated dose is reported twice — 80.9 nSv over the 3,856 s logged against 153.8 nSv integrating across dropouts, 47 % apart — and the program declines to choose.

4 Altitude on airborne tracks

The format carries no altitude, which on an airborne track removes the only variable that explains the count rate. `rcta` imports a trajectory — from the OpenSky Network [5], or a local file offline — and fits an exponential atmosphere (Fig. 2).

The join is the delicate part. *Any* trajectory joins onto *any* track by timestamp, so a wrong flight, day or leg yields a smooth, plausible, entirely fictional curve with no symptom: a fabricated altitude axis is monotonic and the dose really does rise along it. The trajectory's positions are therefore checked against the recorded fixes, and disagreement beyond tolerance is a refusal with its own exit code rather than a plot. Because a clock offset at 233 m/s is indistinguishable from the wrong flight, the offset minimising the residual is reported but never applied automatically. Climb and descent are never pooled: the aircraft crosses the same altitudes twice, hours apart and a continent away, and their disagreement is the honest error bar on the fitted scale height.

5 Limitations and availability

`rcta` performs no source localisation or activity estimation, and tracks carry no spectra, so nuclide identification is impossible from a track alone. Altitude is validated against position, but a trajectory right in position may still be wrong in altitude. The per-sample integration time is inferred from sample spacing because the file does not record it; every detection limit scales directly with it, which makes it the largest systematic in the program, and the tool says so on every run. The jitter correction assumes independent fix errors, so in an urban canyon it is optimistic.

Source, documentation and sample data: `https://github.com/lorenzoconfalonieri/radiacode-track-analyzer`. The software is MIT licensed; one bundled airport table carrying IANA timezones from OurAirports (public domain) and OpenFlights (ODbL) is ODbL 1.0, and analyses made with `rcta` are Produced Works under that licence with no share-alike obligation. Trajectory data are provided by the OpenSky Network [5]; basemap tiles by OpenStreetMap contributors and OpenTopoMap.

References

[1] L. A. Currie, "Limits for qualitative detection and quantitative determination," *Anal. Chem.* **40**, 586–593 (1968).

[2] G. F. Knoll, *Radiation Detection and Measurement*, 4th ed. (Wiley, 2010).

[3] M. S. Bartlett, "On the theoretical specification of sampling properties of autocorrelated time series," *J. R. Stat. Soc. Suppl.* **8**, 27–41 (1946).

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[5] M. Schäfer, M. Strohmeier, V. Lenders, I. Martinovic and M. Wilhelm, "Bringing up OpenSky: A large-scale ADS-B sensor network for research," in *Proc. ACM/IEEE IPSN* (2014), pp. 83–94.